Gradient Descent for Logistic Regression (Multiple Examples)

Syed Hamed Raza

University of Management and Technology (UMT), Lahore
Course: Deep Learning and Neural Network
Course Code: CS4152

November 8, 2025

Recap: Single Example

For one training example (X, Y):

$$z = W^T X + b,$$
$$A = \sigma(z) = \frac{1}{1 + e^{-z}},$$
$$L(A, Y) = -\big[Y \log A + (1 - Y)\log(1 - A)\big].$$

Gradient pieces:

$$\frac{\partial L}{\partial z} = A - Y, \quad \frac{\partial L}{\partial W} = X\,(A - Y), \quad \frac{\partial L}{\partial b} = A - Y.$$

Cost over m Examples

Given m examples $\{(X^{(i)}, Y^{(i)})\}_{i=1}^{m}$, define

$$J(W, b) = \frac{1}{m}\sum_{i=1}^{m} L\big(A^{(i)}, Y^{(i)}\big), \quad A^{(i)} = \sigma\big(W^T X^{(i)} + b\big).$$

We accumulate per-example gradients and then average.

Loop-Based Algorithm (Concept)

Initialize accumulators: $J \leftarrow 0,\ dW \leftarrow 0,\ db \leftarrow 0.$

For each example $i = 1..m$:

$$z^{(i)} = W^T X^{(i)} + b, \qquad A^{(i)} = \sigma\big(z^{(i)}\big),$$
$$J {+}{=} -\Big(Y^{(i)} \log A^{(i)} + (1 - Y^{(i)}) \log(1 - A^{(i)})\Big), \qquad dz^{(i)} = A^{(i)} - Y^{(i)},$$
$$dW {+}{=} X^{(i)}\, dz^{(i)}, \qquad db {+}{=} dz^{(i)}.$$

After the loop: $J \leftarrow J/m,\ dW \leftarrow dW/m,\ db \leftarrow db/m.$

Gradient Descent Update

With learning rate $\alpha > 0$:

$$W \leftarrow W - \alpha \, dW,$$
$$b \leftarrow b - \alpha \, db.$$

One full pass over the m examples constitutes one iteration of gradient descent.

Logistic regression on m examples

$$\underline{J(\omega,b)} = \underbrace{\frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(a^{(i)}, y^{(i)})}$$

$$(x^{(i)}, y^{(i)})$$

$$\rightarrow a^{(i)} = \hat{y}^{(i)} = \sigma(z^{(i)}) = \sigma(\omega^T x^{(i)} + b)$$

$$\underline{d\omega_1^{(i)}}, \underline{d\omega_2^{(i)}}, \underline{db^{(i)}}$$

$$\underline{\frac{\partial}{\partial \omega_1} J(\omega,b)} = \frac{1}{m}\sum_{i=1}^{m} \underbrace{\frac{\partial}{\partial \omega_1} \mathcal{L}(a^{(i)}, y^{(i)})}_{\underline{d\omega_1^{(i)}}\ -\ (x^{(i)},\, y^{(i)})}$$

Gradient Descent on m Examples 02

Logistic regression on m examples

$J=0;\ dw_1=0;\ dw_2=0;\ db=0$

→ For $i=1$ to m

$$z^{(i)} = w^T x^{(i)} + b$$

$$a^{(i)} = \sigma(z^{(i)})$$

$$J += -\left[y^{(i)} \log a^{(i)} + (1-y^{(i)}) \log(1-a^{(i)})\right]$$

$$dz^{(i)} = a^{(i)} - y^{(i)}$$

$$dw_1 \mathrel{+}= x_1^{(i)} dz^{(i)}$$

$$dw_2 \mathrel{+}= x_2^{(i)} dz^{(i)}$$

$$db \mathrel{+}= dz^{(i)}$$

$n=2$

dw_3 ⋮ dw_n

$J /= m$ ←

$dw_1 /= m;\ dw_2 /= m;\ db /= m.$ ←

$$dw_1 = \frac{\partial J}{\partial w_1}$$

$$w_1 := w_1 - \alpha\, dw_1$$

$$w_2 := w_2 - \alpha\, dw_2$$

$$b := b - \alpha\, db$$

Vectorization

A

Vectorized View (Preview)

Stack examples as columns: $X = [X^{(1)}\ X^{(2)}\ \cdots\ X^{(m)}] \in \mathbb{R}^{n\times m}$, labels $Y = [Y^{(1)}, \ldots, Y^{(m)}] \in \mathbb{R}^{1\times m}$.

$$Z = W^T X + b\,\mathbf{1}_{1\times m}, \qquad A = \sigma(Z),$$

$$dZ = A - Y, \qquad dW = \frac{1}{m} X\, dZ^T, \qquad db = \frac{1}{m}\sum_{i=1}^{m} dZ^{(i)}.$$

This removes explicit loops and is far more efficient.

Key Takeaways

Concept	Summary
$dW,\ db$	Average gradients across the m examples
J	Mean of per-example losses
Update	$W \leftarrow W - \alpha dW,\ b \leftarrow b - \alpha db$
Vectorization	Replace loops with matrix ops for speed